

HW#2

TempPub Code :

```
#!/usr/bin/env python3

import rospy
from std_msgs.msg import Float32
import random

def temp_publisher():
    pub = rospy.Publisher('room_temp', Float32, queue_size=10)
    rospy.init_node('TempPub', anonymous=True)
    rate = rospy.Rate(5) # 1 Hz

    while not rospy.is_shutdown():
        temp_celsius = random.uniform(5.0, 30.0)
        rospy.loginfo(f"Publishing Temperature in Celsius:
{temp_celsius:.2f}")
        pub.publish(temp_celsius)
        rate.sleep()

if __name__ == '__main__':
    try:
        temp_publisher()
    except rospy.ROSInterruptException:
        pass
```

FehreSub Code :

```
#!/usr/bin/env python3
import rospy
from std_msgs.msg import Float32

def callback(data):
    fahrenheit = (data.data * 9.0 / 5.0) + 32.0
    rospy.loginfo(f"Temperature in Fahrenheit: {fahrenheit:.2f}")

def fehre_subscriber():
    rospy.init_node('FehreSub', anonymous=True)
    rospy.Subscriber('room_temp', Float32, callback)
    rospy.spin()

if __name__ == '__main__':
    fehre_subscriber()
```

KelvinSub Code :

```
#!/usr/bin/env python3
import rospy
from std_msgs.msg import Float32

def callback(data):
    kelvin = data.data + 273.15
    rospy.loginfo(f"Temperature in Kelvin: {kelvin:.2f}")

def kelvin_subscriber():
    rospy.init_node('KelvinSub', anonymous=True)
    rospy.Subscriber('room_temp', Float32, callback)
    rospy.spin()

if __name__ == '__main__':
    kelvin_subscriber()
```

OUTPUT :

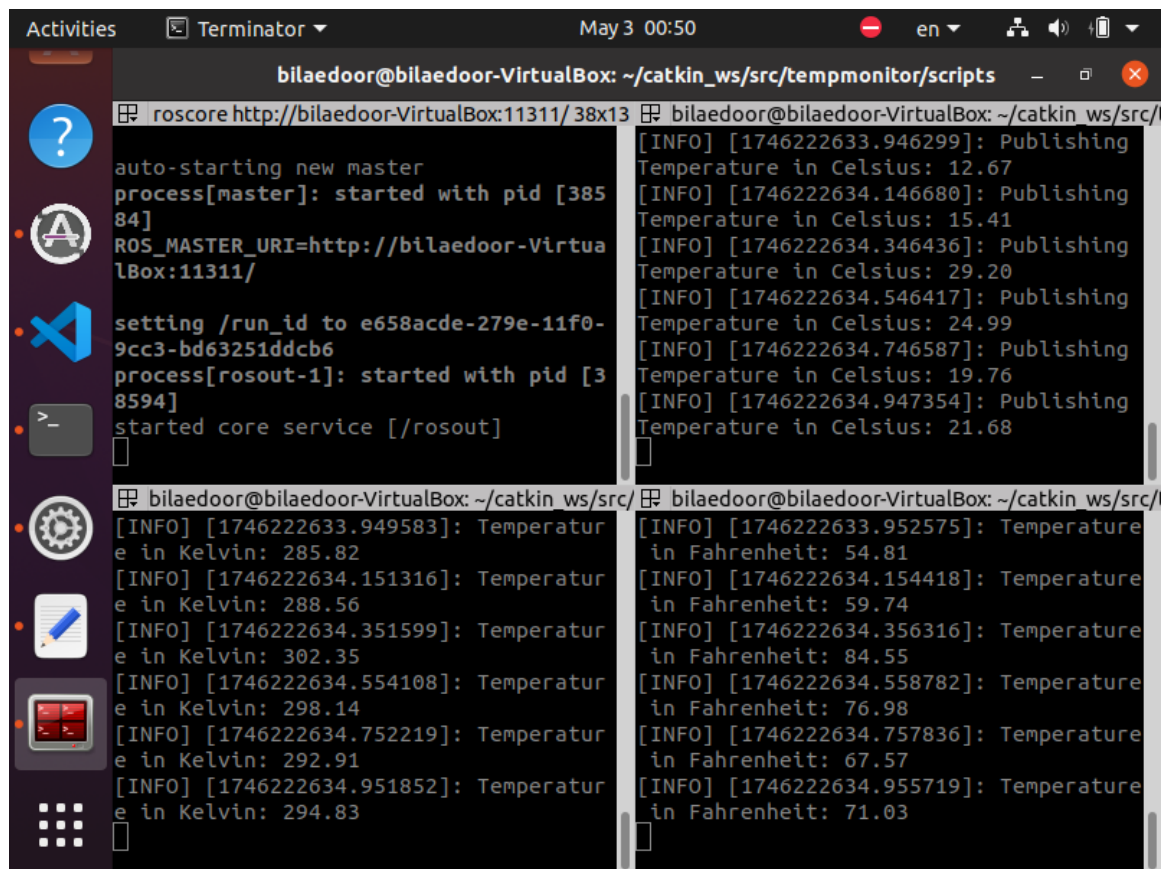
1- Run each node individually using this commands

roscore

roslaunch tempmonitor tempPub.py

roslaunch tempmonitor FehreSub.py

roslaunch tempmonitor KelvinSub.py



The screenshot shows a terminal window titled "bilaedoor@bilaedoor-VirtualBox: ~/catkin_ws/src/tempmonitor/scripts". The terminal displays the output of running ROS nodes. The first part shows the roscore master starting, followed by the tempPub.py node publishing temperature data in Celsius. The second part shows the FehreSub.py node publishing temperature data in Fahrenheit. The third part shows the KelvinSub.py node publishing temperature data in Kelvin. The output is organized into three columns, each representing a different node's output.

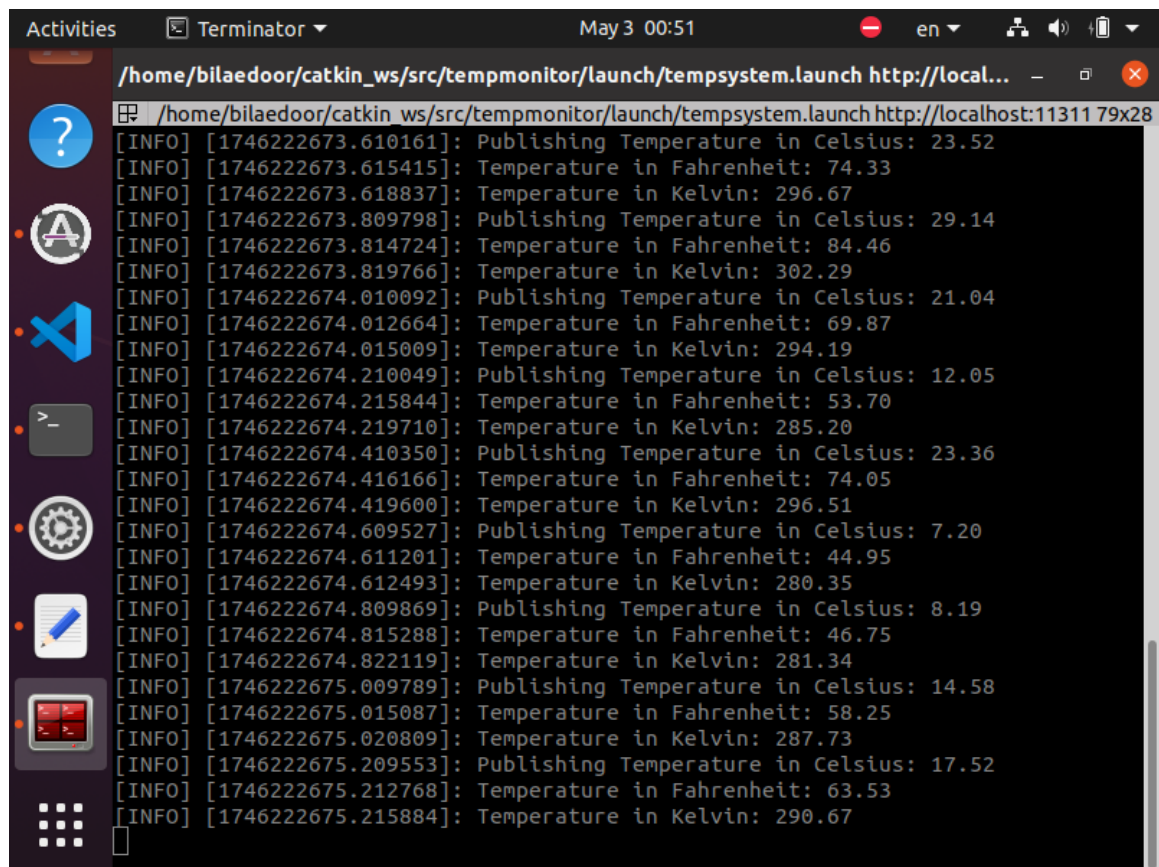
```
bilaeoor@bilaedoor-VirtualBox: ~/catkin_ws/src/tempmonitor/scripts
roscore http://bilaedoor-VirtualBox:11311/ 38x13 bilaeoor@bilaedoor-VirtualBox: ~/catkin_ws/src/
auto-starting new master
process[master]: started with pid [38584]
ROS_MASTER_URI=http://bilaedoor-VirtualBox:11311/
setting /run_id to e658acde-279e-11f0-9cc3-bd63251ddcb6
process[roscout-1]: started with pid [38594]
started core service [/roscout]
[INFO] [1746222633.946299]: Publishing Temperature in Celsius: 12.67
[INFO] [1746222634.146680]: Publishing Temperature in Celsius: 15.41
[INFO] [1746222634.346436]: Publishing Temperature in Celsius: 29.20
[INFO] [1746222634.546417]: Publishing Temperature in Celsius: 24.99
[INFO] [1746222634.746587]: Publishing Temperature in Celsius: 19.76
[INFO] [1746222634.947354]: Publishing Temperature in Celsius: 21.68
bilaeoor@bilaedoor-VirtualBox: ~/catkin_ws/src/ bilaeoor@bilaedoor-VirtualBox: ~/catkin_ws/src/
[INFO] [1746222633.949583]: Temperature in Kelvin: 285.82
[INFO] [1746222634.151316]: Temperature in Kelvin: 288.56
[INFO] [1746222634.351599]: Temperature in Kelvin: 302.35
[INFO] [1746222634.554108]: Temperature in Kelvin: 298.14
[INFO] [1746222634.752219]: Temperature in Kelvin: 292.91
[INFO] [1746222634.951852]: Temperature in Kelvin: 294.83
bilaeoor@bilaedoor-VirtualBox: ~/catkin_ws/src/ bilaeoor@bilaedoor-VirtualBox: ~/catkin_ws/src/
[INFO] [1746222633.952575]: Temperature in Fahrenheit: 54.81
[INFO] [1746222634.154418]: Temperature in Fahrenheit: 59.74
[INFO] [1746222634.356316]: Temperature in Fahrenheit: 84.55
[INFO] [1746222634.558782]: Temperature in Fahrenheit: 76.98
[INFO] [1746222634.757836]: Temperature in Fahrenheit: 67.57
[INFO] [1746222634.955719]: Temperature in Fahrenheit: 71.03
```

1- Using Launch file

Using this Command :

Roslaunch tempmonitor tempsystem.launch

```
<launch>
  <node pkg="tempmonitor" type="tempPub.py" name="publisher_node"
output="screen" />
  <node pkg="tempmonitor" type="fehreSub.py" name="fahrenheit_node"
output="screen" />
  <node pkg="tempmonitor" type="kelvinSub.py" name="kelvin_node"
output="screen" />
</launch>
```



```
/home/bilaedoor/catkin_ws/src/tempmonitor/launch/tempsystem.launch http://local...
/home/bilaedoor/catkin_ws/src/tempmonitor/launch/tempsystem.launch http://localhost:11311 79x28
[INFO] [1746222673.610161]: Publishing Temperature in Celsius: 23.52
[INFO] [1746222673.615415]: Temperature in Fahrenheit: 74.33
[INFO] [1746222673.618837]: Temperature in Kelvin: 296.67
[INFO] [1746222673.809798]: Publishing Temperature in Celsius: 29.14
[INFO] [1746222673.814724]: Temperature in Fahrenheit: 84.46
[INFO] [1746222673.819766]: Temperature in Kelvin: 302.29
[INFO] [1746222674.010092]: Publishing Temperature in Celsius: 21.04
[INFO] [1746222674.012664]: Temperature in Fahrenheit: 69.87
[INFO] [1746222674.015009]: Temperature in Kelvin: 294.19
[INFO] [1746222674.210049]: Publishing Temperature in Celsius: 12.05
[INFO] [1746222674.215844]: Temperature in Fahrenheit: 53.70
[INFO] [1746222674.219710]: Temperature in Kelvin: 285.20
[INFO] [1746222674.410350]: Publishing Temperature in Celsius: 23.36
[INFO] [1746222674.416166]: Temperature in Fahrenheit: 74.05
[INFO] [1746222674.419600]: Temperature in Kelvin: 296.51
[INFO] [1746222674.609527]: Publishing Temperature in Celsius: 7.20
[INFO] [1746222674.611201]: Temperature in Fahrenheit: 44.95
[INFO] [1746222674.612493]: Temperature in Kelvin: 280.35
[INFO] [1746222674.809869]: Publishing Temperature in Celsius: 8.19
[INFO] [1746222674.815288]: Temperature in Fahrenheit: 46.75
[INFO] [1746222674.822119]: Temperature in Kelvin: 281.34
[INFO] [1746222675.009789]: Publishing Temperature in Celsius: 14.58
[INFO] [1746222675.015087]: Temperature in Fahrenheit: 58.25
[INFO] [1746222675.020809]: Temperature in Kelvin: 287.73
[INFO] [1746222675.209553]: Publishing Temperature in Celsius: 17.52
[INFO] [1746222675.212768]: Temperature in Fahrenheit: 63.53
[INFO] [1746222675.215884]: Temperature in Kelvin: 290.67
```

Rqt_graph:

